



# *Towards Unification of Name and Address Based Communication*

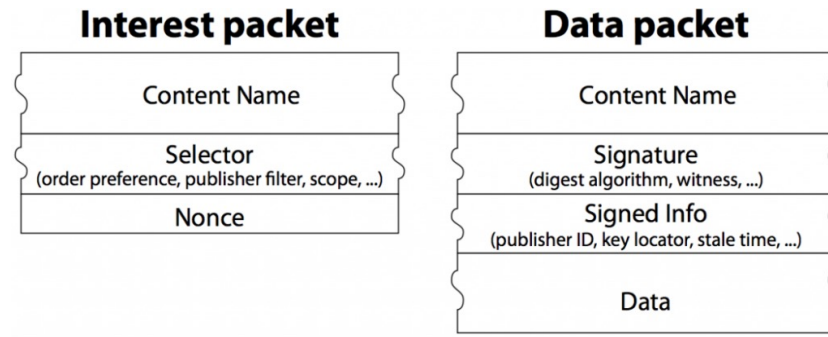
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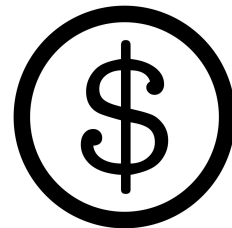
# Motivation

NFD has been out for multiple years



**NAMED DATA  
NETWORKING**

Industry hasn't adopted NFD yet due to decades of vested interest and TCP/IP works



**NFD needs a success story where address-based stack  
cannot compete**

# *Where Name-based Can Excel*

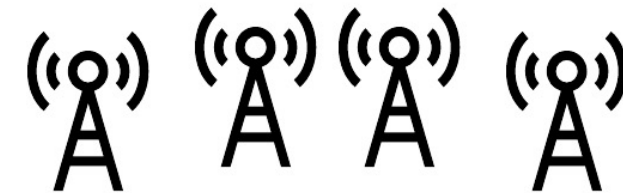
## Decentralized wild edge

Standard procedures are still in the formation process



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Edge name-based multicast dissemination is needed



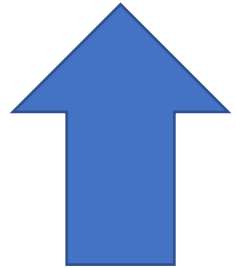
one to many communication



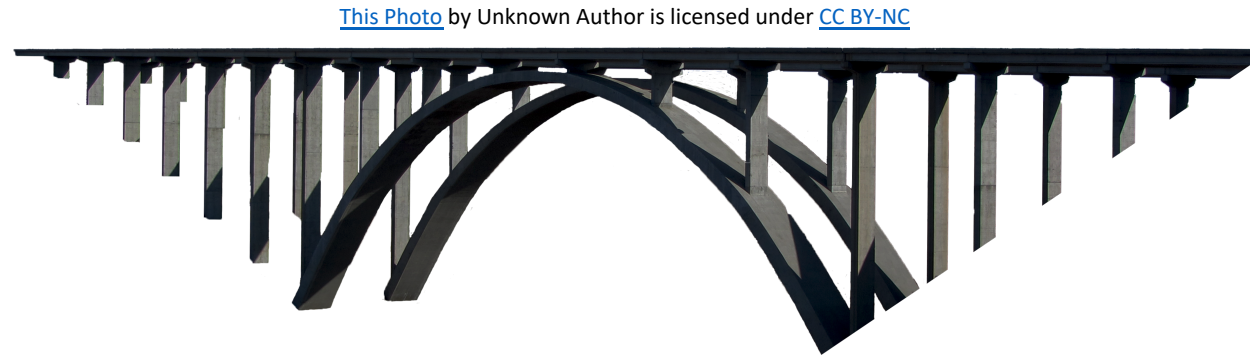
# How Do We Get Industry Attention?



Address-based



Current Industry  
status



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TCP/IP with lightweight name-  
based services

Unified  
Paradigm



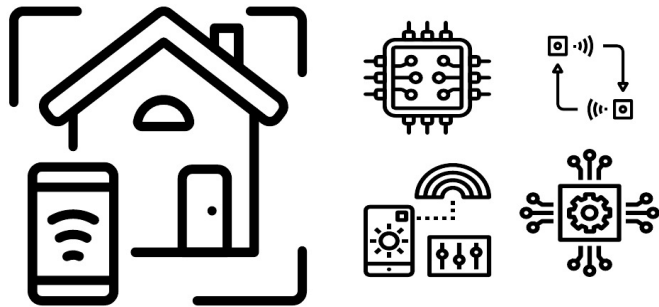
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**e.g., Data multicast dissemination on the edge**  
*Name\_stack\_distribute(file, security\_cred)*

# *Why Unified Instead of Name-based*



Access Control is commands-driven  
*i.e., transmitter initiates (ID/address)*



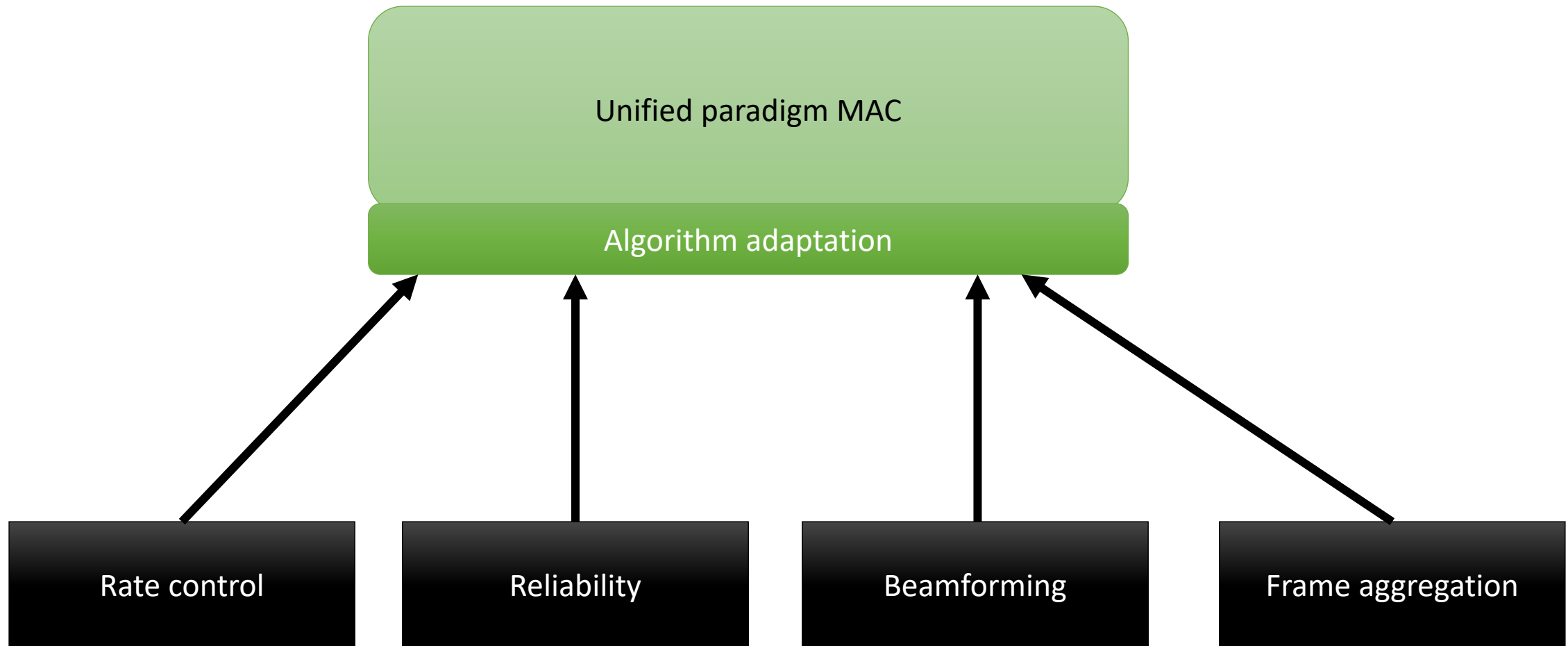
***Access Control***

Applications are multicast data driven  
*i.e., receiver initiates (name-based)*



***Augmented Vehicular Reality***

# *Unified Paradigm Stack*



# What is Rate Control



A

## What data rate to transmit at?



B

Data rates sample table

| MCS index | Modulation | Code rate | Spatial streams | FEC coders | PHY rate |        |
|-----------|------------|-----------|-----------------|------------|----------|--------|
|           |            |           |                 |            | 20 MHz   | 40 MHz |
| 0         | BPSK       | 1/2       | 1               | 1          | 6.5      | 13.5   |
| 1         | QPSK       | 1/2       | 1               | 1          | 13       | 27     |
| 7         | 64-QAM     | 5/6       | 1               | 1          | 65       | 135    |
| 8         | BPSK       | 1/2       | 2               | 1          | 13       | 27     |
| 14        | 64-QAM     | 3/4       | 2               | 1          | 117      | 243    |
| 21        | 64-QAM     | 2/3       | 2               | 2          | 156      | 324    |
| 28        | 16-QAM     | 3/4       | 4               | 2          | 156      | 324    |
| 31        | 64-QAM     | 5/6       | 4               | 2          | 260      | 540    |
| 31*       | 64-QAM     | 5/6       | 4               | 2          | 288.89   | 600    |

Adjustment is per-frame based on medium condition that changes dynamically.

# How Does Rate Control Work



Supported Rates  
between A and B

| Data Rate<br>(Mbps) |
|---------------------|
| 1                   |
| 6                   |
| 9                   |
| 24                  |
| 54                  |

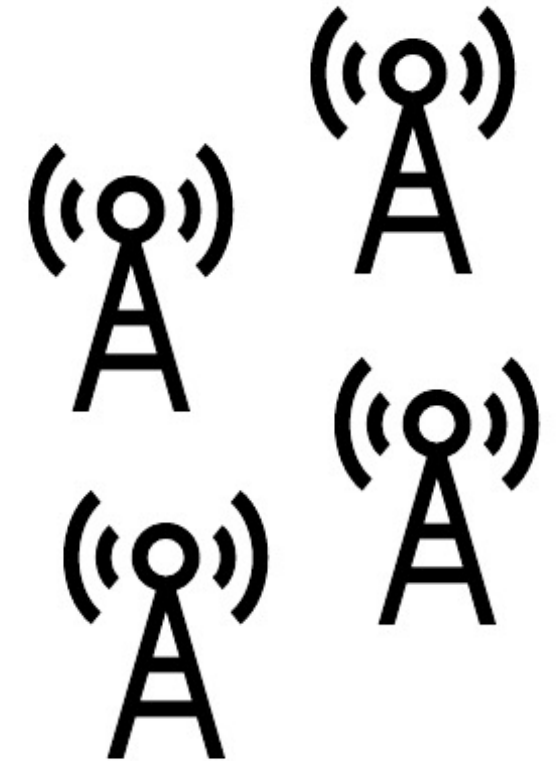
## Scenario Walkthrough:

F1 sent at 9 mbps ----- acknowledged  
 F2 sent at 24Mbps ----- **lost**  
 F2 sent at 9Mbps ----- acknowledged  
 F3 sent at 9Mbps ----- acknowledged  
 F4 sent at 24Mbps ----- acknowledged

# What About Name-based Rate Control?



How do we adapt  
data rate based on  
unknown receivers?

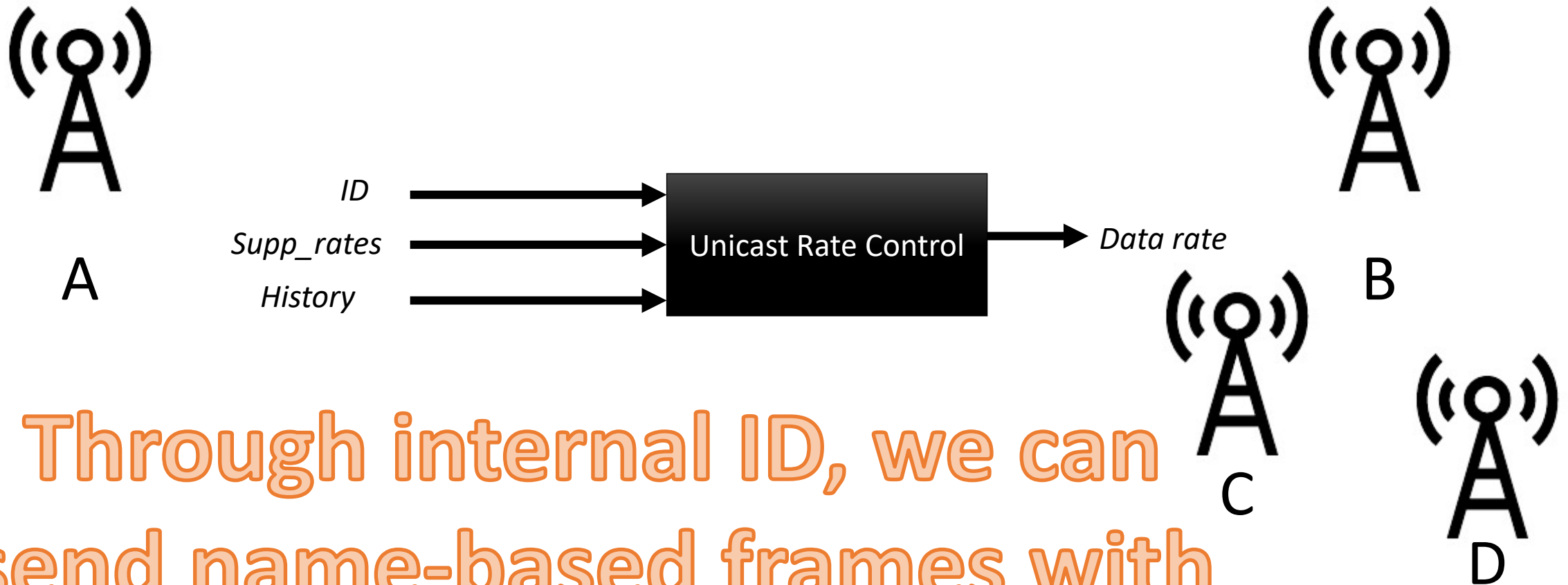


*We can restart rate control every new packet.*

*However, the results are poor due to limited number of frames before reset.*

Embed ID information per  
Interest to allow rate Control

# Unified MAC Rate Control



Through internal ID, we can  
send name-based frames with  
rate control

# *Preliminary Results from Unified MAC*



*Setup:* **Single consumer with prior transmission Results**

| Mode                       | Latency (s) | Loss Rate (%) | Medium Utilization (bandwidth) (%) |
|----------------------------|-------------|---------------|------------------------------------|
| Broadcast                  | 30          | 12 – 20       | 35                                 |
| Unified Stack transmission | 0.9         | 0             | 1-3                                |

**Rate Control algorithm is critical for  
good performance in wireless  
communication**



# *Existing Assets*

- Filtering
  - Name and address-based
- Discovery
  - Attribute-based discovery for address-based communication
- Rate Control
  - Unicast
- Multicast robustness protocol
- Single consumer producer selection
- Frame aggregation
- Beamforming



# *Needs to Be Done*

- Multicast Rate Control
- Multi-consumer producer selection



# *Conclusions*

- Attract Industry Professionals by targeting edge market
- Proposal for lightweight name-based services marketing for low barrier entry and immediate benefit
- Unified Medium Access Control (MAC) that benefits both name and address-based communications



# *Thank You Questions?*

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